

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. Examination, 2020
CSE – 1102, Discrete Mathematics

Total Marks: 70

Time: 2 Hours

(Answer any 3 (Three) of the following Questions)

1. a) Define Set with example. Prove that $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$, where A_1 and A_2 be sets with cardinalities $|A_1|$ and $|A_2|$. 2+
5.33
 b) Explain Venn Diagram for Union and Intersection operations where A and B are two sets. 6
 c) Write down the Laws of set theory. 4
 d) Consider Universal set U and two sets A and B. Shade the following sets using Venn Diagram: 6

i) $A \cap B^C$
ii) $(B/A)^C$

2. a) What is recursion? A function f is defined recursively by $f(0) = 3$, $f(n+1) = 2f(n)+3$; Find the value of $f(4)$. 3+3
 b) What is recurrence relation? Obtain the recurrence relation for:

$$G(k) = 2.4^k - 5.(-3)^k$$
3+6
 c) Find the solution to the recurrence relation: $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ with the initial conditions $a_0=2$, $a_1=5$ and $a_2=15$ 8.33

3. a) Define reflexive, symmetric and transitive relation. 6
 b) Consider the following five relations on the set $A = \{1, 2, 3, 4\}$: 9
 $R = \{(1, 1), (1, 2), (2, 3), (1, 3), (4, 4)\}$
 $S = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$
 $T = \{(1, 3), (2, 1)\}$
 $U = A \times A$, the universal relation
 $V = \emptyset$, the empty relation
 Determine whether or not each of the above relations on A is
 i) Reflexive ii) Symmetric iii) Transitive
 c) Define Equivalence relation with example. 3.33
 d) Let $A = \{1,2,3,4\}$, $B = \{4,5,6\}$ and $C = \{6,7,8\}$. R is a relation from A to B and S is a relation from B to C, which is given by

$$R = \{(x,y): x + y = 7\}$$

$$S = \{(x,y): y - x = 1\}$$
 Determine R and S.

4. a) State the converse, contrapositive and inverse of the following conditional statements: 6
 i) If it snows tonight, then I will stay at home.
 ii) I go to the beach whenever it is a sunny summer day.
 b) State and prove the Principle of Inclusion and Exclusion. 6
 c) Let $A = \{1,2,3\}$ and R be the relation, $R = \{(1,2), (1,3), (2,2), (2,3), (3,3)\}$. Draw the diagram by using this relation. 4
 d) Define tautology and contradiction with examples. Verify that $(p \wedge q) \wedge \neg (p \vee q)$ is a contradiction by using truth table. 4+
3.33

5. a) Define with figure:

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- i) Graph
- ii) Subgraph
- iii) Bipartite graph
- iv) Complete graph
- v) Directed graph

b) Draw the graph G corresponding to the adjacency matrix:

3.33

$$A = \begin{bmatrix} 1 & 3 & 0 & 0 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix}$$

c) What is isomorphism? Draw two graphs that are isomorphic and describe briefly.

2+4

d) Represent the graph with an incidence matrix.

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